This modified scenario is relevant to cosmology because, as we've learned, like the pranksters' rubber bands, a uniform inflaton field exerts a negative pressure within an exanding universe. And so, just as the total energy embodied by the rubber bands increases as the box expands because they extract energy from the box's walls, the total energy embodied by the inflaton field increases as the universe expands because it *extracts energy from gravity.*[27]

To summarize: *as the universe expands, matter and radiation lose energy to gravity while an inflaton field gains energy from gravity.*

The pivotal nature of these observations becomes clear when we try to explain the origin of the matter and radiation that make up galaxies, stars, and everything else inhabiting the cosmos. In the standard big bang theory, the mass/energy carried by matter and radiation has steadily decreased as the universe has expanded, and so the mass/energy in the early universe greatly exceeded what we see today. Thus, instead of offering an explanation for where all the mass/energy currently inhabiting the universe originated, the standard big bang fights an unending uphill battle: the farther back the theory looks, the *more* mass/energy it must somehow explain.

In inflationary cosmology, though, much the opposite is true. Recall that the inflationary theory argues that matter and radiation were produced at the end of the inflationary phase as the inflaton field released its pent-up energy by rolling from perch to valley in its potential-energy bowl. The relevant question, therefore, is whether, just as the inflationary phase was drawing to a close, the theory can account for the inflaton field embodying the *stupendous* quantity of mass/energy necessary to yield the matter and radiation in today's universe.

The answer to this question is that inflation can, without even breaking a sweat. As just explained, the inflaton field is a gravitational parasite—it feeds on gravity—and so the total energy the inflaton field carried increased as space expanded. More precisely, the mathematical analysis shows that the energy *density* of the inflaton field remained constant throughout the inflationary phase of rapid expansion, implying that the total energy it embodied grew in direct proportion to the volume of the space it filled. In the previous chapter, we saw that the size of the